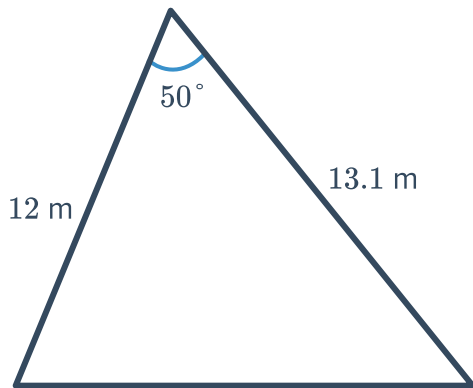


Area of non-right triangles

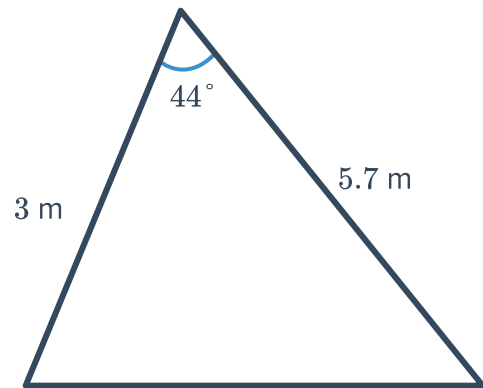


1 Calculate the area of the following triangles, rounded to two decimal places:

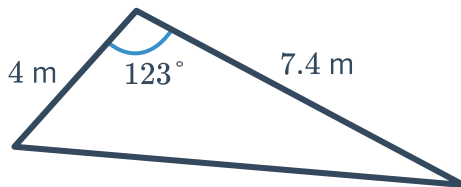
a



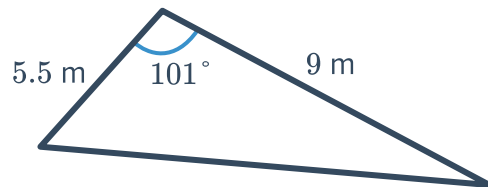
b



c



d



2 Find the area of the following triangles, to one decimal place:

a Sides lengths of 9.5 ft, 10 ft with an included angle measuring 47° .

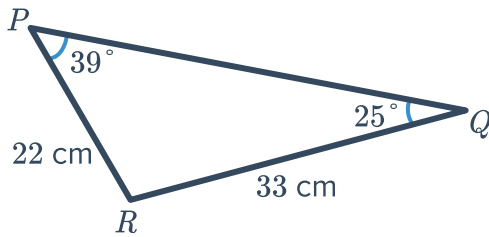
b Sides lengths of 18.4 in, 20.5 in with an included angle measuring 99° .

3 Find the exact area of an equilateral triangle with a side length of 6 yd.

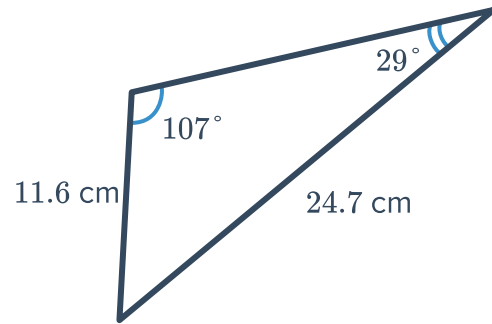
4 Find the exact area of an equilateral triangle with a perimeter of 24 in.

5 Calculate the area of the following triangles to the nearest square centimeter:

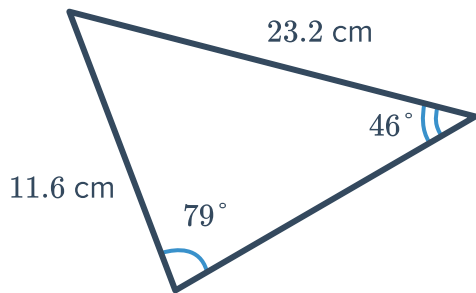
a



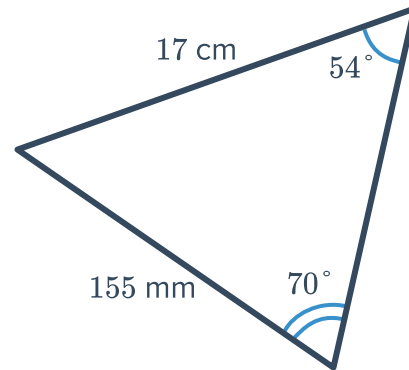
b



c

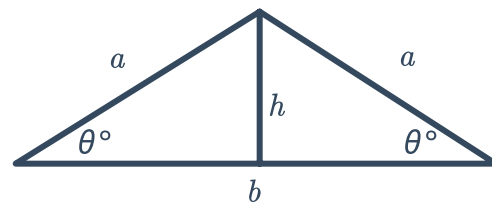


d

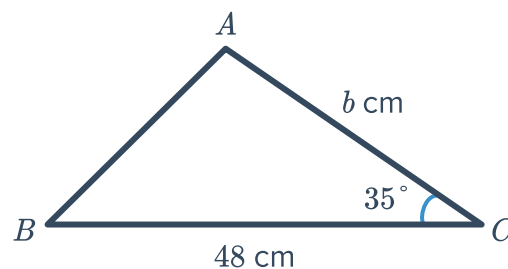


6 Consider the diagram of an isosceles triangle where h is the height perpendicular to base, b :

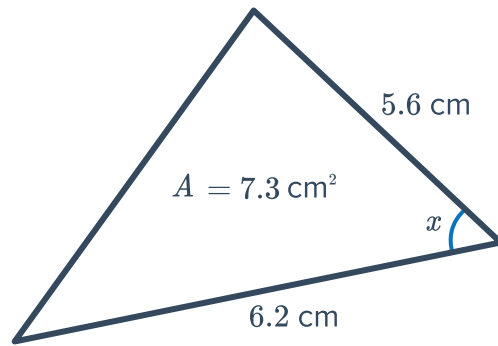
- a Form an expression for h in terms of θ and a .
- b Find b in terms of θ and a .
- c Form an expression for the area A of the larger triangle, in terms of θ and a .



7 The following triangle has an area of 520 cm^2 . Find the length of side b . Round your answer to the nearest centimeter.



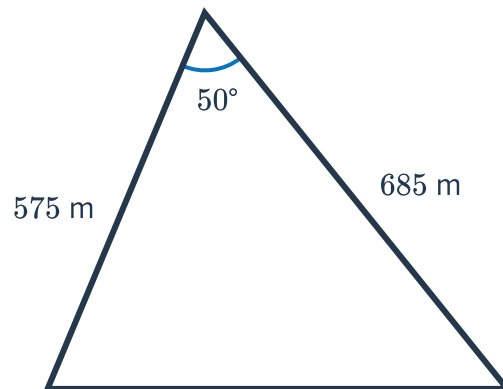
- 8 Consider the diagram and find the measure of angle x , correct to one decimal place:



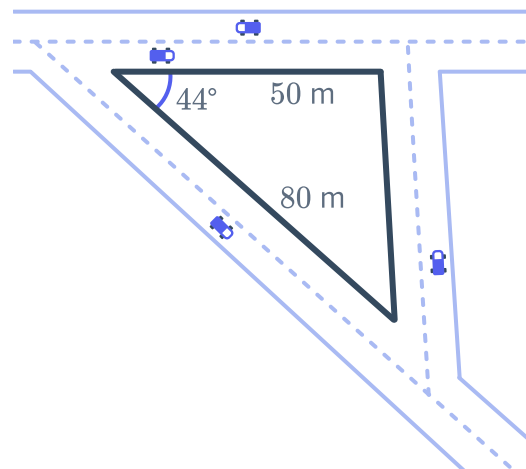
Applications

- 9 A triangular paddock has measurements as shown in the diagram:

- a Find the area of the paddock. Round your answer to the nearest square meter.
- b State the area in hectares. Round your answer to two decimal places.



- 10 An industrial site in the shape of a triangle is to take up the space between where three roads intersect as shown in the diagram. Find the area of the site. Round your answer to two decimal places.



- 11 The Bermuda Triangle is an area in the Atlantic Ocean where many planes and ships have mysteriously disappeared. Its vertices are at Bermuda (B), Miami (M) and Puerto Rico (P):



Find the area of the Bermuda Triangle. Round your answer correct to two decimal places.